# Mathematical Modeling

Sheet 1: Dual Simplex and Primal Recovery

Tasks 4 and 5(a)–5(d)

Personal study notes

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## 1 Task statements

**4.** Rozwiąż algorytmem dualnym. Na tej podstawie podaj rozwiązanie zagadnienie dualnego.

$$\begin{aligned} \min \quad & x_1 + 6x_2 + 2x_3 + 3x_4, \\ & 4x_1 + 2x_2 - x_3 + x_4 \geq 4, \\ & x_1 + x_2 + x_3 + x_4 \geq 5, \\ & -2x_1 - 2x_2 - x_3 - x_4 \geq -8, \\ & x_1, x_2, x_3, x_4 \geq 0. \end{aligned}$$

**5.**

- Podaj postać zadania dualnego ZD
- Rozwiąż ZD dwufazową metodą sympleks i podaj jego znalezione RO oraz optymalną wartość funkcji celu dla ZD
- Na podstawie znalezionego rozwiązania RO dla zadania ZD, oblicz RO dla zadania ZP ( bez rozwiązywania ZP) oraz sprawdź optymalną wartość funkcji w ZP
- Czy zadanie ZD posiada rozwiązania alternatywne? Uzasadnij

**(a)**

$$\begin{aligned} \min \quad & 3x_1 + 10x_2 + 2x_3, \\ & -x_1 + 2x_2 + x_3 \geq 2, \\ & -x_1 + x_2 + x_3 \geq 1, \\ & -x_1 + 2x_2 - x_3 \geq -3, \\ & -x_1 + 2x_2 + x_3 \geq 1, \\ & x_1 \leq 0, \quad x_2 \geq 0, \quad x_3 \geq 0. \end{aligned}$$

**(b)**

$$\begin{aligned} \min \quad & 3x_1 + 10x_2 + x_3, \\ & -2x_2 - 2x_3 \geq -2, \\ & 2x_1 - x_2 \geq 1, \\ & -2x_1 + x_2 + x_3 \geq -5, \\ & 2x_1 + 2x_2 + x_3 \geq 1, \\ & x_1 \geq 0, \quad x_2 \in \mathbb{R}, \quad x_3 \geq 0. \end{aligned}$$

**(c)**

$$\begin{aligned} \min \quad & -x_1 + 5x_2 + 8x_3, \\ & -x_1 - x_2 + 2x_3 \geq -1, \\ & -x_1 + 3x_3 \geq 5, \\ & -x_1 + x_2 + x_3 \geq 1, \\ & -x_2 - 2x_3 \geq -1, \\ & x_1 \in \mathbb{R}, \quad x_2 \in \mathbb{R}, \quad x_3 \geq 0. \end{aligned}$$

**(d)**

$$\begin{aligned} \min \quad & x_1 - 5x_2 + 8x_3, \\ & 3x_1 + x_2 + 2x_3 \geq 1, \\ & -x_1 + x_3 \geq 5, \\ & -x_1 - x_2 + x_3 \geq 1, \\ & 3x_1 - x_2 - 2x_3 \geq -2, \\ & x_1 \geq 0, \quad x_2 \in \mathbb{R}, \quad x_3 \geq 0. \end{aligned}$$

## 2 Notation, domains, and the mathematical plan

All decision variables are continuous and obey the displayed sign restrictions. The data specify abstract linear programs, with no physical units supplied. In particular, a free variable may be negative: it must not acquire an extra nonnegativity restriction during conversion or verification.

The prerequisites are standard-form equivalence and bases in Chapter 02, Phase I and reduced costs in Chapter 03, and mixed-domain duality, complementarity, and dual simplex in Chapter 04. We apply these ideas to the complete tasks below.

For any equality-form maximization, use

$$\max c^T u, \qquad Mu = d, \quad u \geq 0.$$

A nonsingular basis $B$ gives

$$\bar{M} = M_B^{-1} M, \quad \bar{d} = M_B^{-1} d, \quad \lambda = M_B^{-T} c_B, \quad q = M^T \lambda - c.$$

The tableau displays the equations $\bar{M}u = \bar{d}$ and the last row $q^T u = z_0 - c^T u$, where $z_0 = c_B^T \bar{d}$. Thus

$$c^T u = z_0 - \sum_{j \notin B} q_j u_j. \tag{1}$$

The last RHS is $z_0$, not a basic variable. Nonnegative indicators certify an upper bound on the maximum. The original minimum in task 4 is the negative of its transformed maximum. Tasks 5 maximize their dual objectives directly.

## 2.1 Why the domains determine the dual inequalities

Write a primal minimum as $\min c^T x$, $Ax \geq b$. A nonnegative multiplier $y$ gives $y^T Ax \geq y^T b$. To make $c^T x \geq y^T Ax$ hold for every allowed $x$, column $j$ must obey

| Primal domain | Dual condition |
|---|---|
| $x_j \geq 0$ | $a_j^T y \leq c_j$ |
| $x_j \leq 0$ | $a_j^T y \geq c_j$ |
| $x_j \in \mathbb{R}$ | $a_j^T y = c_j$ |

For example, multiplying a nonpositive $x_j$ by $c_j - a_j^T y \leq 0$ gives a nonnegative product. A free variable can take either sign, so its coefficient must vanish. The resulting dual maximizes $b^T y$ over $y \geq 0$ with these column conditions. Negative entries in $b$ do not change multiplier domains.

For a feasible pair, the gap has the decomposition

$$c^T x - b^T y = (c - A^T y)^T x + y^T (Ax - b) \geq 0. \tag{2}$$

Every term on the right is nonnegative with the appropriate domain convention. Optimal pairs make each term zero. This explains complementary slackness, rather than treating it as an additional system to memorize.

## 2.2 Two different simplex invariants

Primal simplex keeps $\bar{d} \geq 0$. It chooses an entering column with $q_k < 0$ and preserves feasibility by the ratio test

$$\theta = \min_{\bar{m}_{ik} > 0} \frac{\bar{d}_i}{\bar{m}_{ik}}.$$

Our Phase I and II paths choose the most negative indicator, breaking ties by smaller column index, then the smallest ratio, breaking ties by smaller leaving variable index. These choices specify these finite paths; they are not a general anti-cycling guarantee.

Dual simplex keeps $q \geq 0$. It first selects a negative RHS row $r$. Eligible entering columns have $\bar{m}_{rk} < 0$, and

$$\alpha = \min_{\bar{m}_{rj} < 0} \frac{q_j}{-\bar{m}_{rj}}$$

preserves nonnegative indicators because $q'_j = q_j + \alpha \bar{m}_{rj}$. We select the most negative RHS and the smaller entering column in a tie. The update $z'_0 = z_0 + \alpha \bar{d}_r$ lowers the dual upper bound on a maximum. It does not describe improvement of an infeasible primal point.

Phase I maximizes minus the sum of nonnegative artificial variables. Its optimum is at most zero. A negative optimum certifies infeasibility; zero permits Phase II after artificials are removed and the original objective is recomputed. All successful paths here remove artificials from the basis, so no zero-basic artificial cleanup is needed.

## 3 Task 4: a complete dual-simplex solve

Put $s_i = (Ax - b)_i \geq 0$, negate each row, and maximize the negative cost:

$$\begin{aligned} &\max \quad -x_1 - 6x_2 - 2x_3 - 3x_4, \\ -4x_1 - 2x_2 + x_3 - x_4 + s_1 &= -4, \\ -x_1 - x_2 - x_3 - x_4 + s_2 &= -5, \\ 2x_1 + 2x_2 + x_3 + x_4 + s_3 &= 8. \end{aligned}$$

The surplus basis $B = (s_1, s_2, s_3)$ is nonsingular, with indicators $(1, 6, 2, 3, 0, 0, 0) \geq 0$. Its negative RHS entries make it primal infeasible, but exactly the dual-feasible start we need.

| Basic | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $s_1$ | $s_2$ | $s_3$ | RHS |
|---|---|---|---|---|---|---|---|---|
| $s_1$ | $-4$ | $-2$ | 1 | $-1$ | 1 | 0 | 0 | $-4$ |
| $s_2$ | $-1$ | $-1$ | $-1$ | $-1$ | 0 | 1 | 0 | $-5$ |
| $s_3$ | 2 | 2 | 1 | 1 | 0 | 0 | 1 | 8 |
| $q^T$ | 1 | 6 | 2 | 3 | 0 | 0 | 0 | 0 |

The most negative RHS is $-5$: $s_2$ leaves. The eligible columns $x_1, x_2, x_3, x_4$ have ratios $1, 6, 2, 3$, respectively, so $x_1$ enters on the pivot $-1$. Divide row 2 by $-1$, then eliminate column 1 from rows 1 and 3. The new bound is $z_0 = 0 + 1(-5) = -5$.

| Basic | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $s_1$ | $s_2$ | $s_3$ | RHS |
|---|---|---|---|---|---|---|---|---|
| $s_1$ | 0 | 2 | 5 | 3 | 1 | $-4$ | 0 | 16 |
| $x_1$ | 1 | 1 | 1 | 1 | 0 | $-1$ | 0 | 5 |
| $s_3$ | 0 | 0 | $-1$ | $-1$ | 0 | 2 | 1 | $-2$ |
| $q^T$ | 0 | 5 | 1 | 2 | 0 | 1 | 0 | $-5$ |

Row 3 still has RHS $-2$, so $s_3$ leaves. Only $x_3$ and $x_4$ are eligible; their ratios are $1/1 = 1$ and $2/1 = 2$. Enter $x_3$ on the pivot $-1$. The next bound is $-5 + 1(-2) = -7$.

| Basic | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $s_1$ | $s_2$ | $s_3$ | RHS |
|---|---|---|---|---|---|---|---|---|
| $s_1$ | 0 | 2 | 0 | $-2$ | 1 | 6 | 5 | 6 |
| $x_1$ | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 3 |
| $x_3$ | 0 | 0 | 1 | 1 | 0 | $-2$ | $-1$ | 2 |
| $q^T$ | 0 | 5 | 0 | 1 | 0 | 3 | 1 | $-7$ |

Now both invariants hold: every RHS and indicator is nonnegative. Reading nonbasic variables as zero gives

$$x^* = (3, 0, 2, 0), \quad s^* = (6, 0, 0), \qquad \min c^T x = 7.$$

The final tableau also gives the exact identity

$$c^T x = 7 + 5x_2 + x_4 + 3s_2 + s_3 \geq 7. \tag{3}$$

Equality forces $x_2 = x_4 = s_2 = s_3 = 0$. The last two tight primal rows then give $x_1 + x_3 = 5$ and $2x_1 + x_3 = 8$, so this primal optimizer is unique.

### 3.1 Read and verify the dual certificate

The original minimum has dual

$$\begin{aligned} \max \quad & 4y_1 + 5y_2 - 8y_3, \\ 4y_1 + y_2 - 2y_3 &\leq 1, \\ 2y_1 + y_2 - 2y_3 &\leq 6, \\ -y_1 + y_2 - y_3 &\leq 2, \\ y_1 + y_2 - y_3 &\leq 3, \qquad y \geq 0. \end{aligned}$$

For the transformed equations $[-A, I](x, s) = -b$, the basis multiplier satisfies $-A^T y \geq -c$ and $y \geq 0$, which are precisely these constraints. The last tableau's original surplus indicators give

$$y^* = (0, 3, 1), \quad c - A^T y^* = (0, 5, 0, 1), \quad b^T y^* = 15 - 8 = 7.$$

We have primal and dual feasibility, equal values, and $y^* \odot s^* = 0$, $x^* \odot (c - A^T y^*) = 0$. To see dual uniqueness, $s_1 = 6 > 0$ forces $y_1 = 0$ in every dual optimum. The positive $x_1$ and $x_3$ force $y_2 - 2y_3 = 1$ and $y_2 - y_3 = 2$, whose unique solution is $(y_2, y_3) = (3, 1)$.

## 4 Task 5(a): Phase I proves dual infeasibility

The nonpositive domain of $x_1$ reverses its dual inequality. The dual is

$$\begin{aligned} \max \quad & 2y_1 + y_2 - 3y_3 + y_4, \\ -y_1 - y_2 - y_3 - y_4 &\geq 3, \\ 2y_1 + y_2 + 2y_3 + 2y_4 &\leq 10, \\ y_1 + y_2 - y_3 + y_4 &\leq 2, \qquad y \geq 0. \end{aligned}$$

Its first condition is impossible because its left side is nonpositive. We can also express this contradiction through the requested two-phase procedure. Introduce surplus $t_1$ and artificial $a_1$ in row 1, and slacks $s_2, s_3$:

$$-y_1 - y_2 - y_3 - y_4 - t_1 + a_1 = 3.$$

Use $B = (a_1, s_2, s_3)$ and maximize $-a_1$. The other two equations have right sides 10 and 2.

| Basic | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $t_1$ | $s_2$ | $s_3$ | $a_1$ | RHS |
|---|---|---|---|---|---|---|---|---|---|
| $a_1$ | −1 | −1 | −1 | −1 | −1 | 0 | 0 | 1 | 3 |
| $s_2$ | 2 | 1 | 2 | 2 | 0 | 1 | 0 | 0 | 10 |
| $s_3$ | 1 | 1 | −1 | 1 | 0 | 0 | 1 | 0 | 2 |
| $q^T$ | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | −3 |

Every indicator is already nonnegative. Therefore Phase I has optimum $-3$, or minimum artificial sum $w = 3$. More directly,

$$a_1 = 3 + y_1 + y_2 + y_3 + y_4 + t_1 \geq 3.$$

A feasible original dual would need $a_1 = 0$, which is impossible. Stop before Phase II: there is no dual optimum and hence no optimal pair to recover.

### 4.1 A primal ray explains the same outcome

The ray

$$x(t) = (-2 - t, 0, 0), \qquad t \geq 0$$

respects all domains and has primal row values $(2+t, 2+t, 2+t, 2+t)$, which dominate $(2, 1, -3, 1)$. Its objective is $-6 - 3t \to -\infty$. Thus the primal is feasible and unbounded below. This is consistent with the dual contradiction: a feasible dual would supply a finite lower bound and would forbid such a ray. Neither problem has an optimal solution; "alternative optimal dual solutions" therefore do not exist.

## 5 Task 5(b): solve the dual and recover the primal

Because $x_2$ is free, its column gives an equality:

$$\begin{aligned} \max \quad & -2y_1 + y_2 - 5y_3 + y_4, \\ 2y_2 - 2y_3 + 2y_4 &\leq 3, \\ -2y_1 - y_2 + y_3 + 2y_4 &= 10, \\ -2y_1 + y_3 + y_4 &\leq 1, \qquad y \geq 0. \end{aligned}$$

Let $s_1, s_3$ be the slacks of the two dual inequalities, and add $a_2$ to its equality. Column order is $(y_1, y_2, y_3, y_4, s_1, s_3, a_2)$. Notice that these slacks index dual conditions, not primal resource rows.

### 5.1 Phase I: three pivots remove the artificial variable

Maximize $-a_2$, starting at $B = (s_1, a_2, s_3)$.

| Basic | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $s_1$ | $s_3$ | $a_2$ | RHS |
|---|---|---|---|---|---|---|---|---|
| $s_1$ | 0 | 2 | −2 | 2 | 1 | 0 | 0 | 3 |
| $a_2$ | −2 | −1 | 1 | 2 | 0 | 0 | 1 | 10 |
| $s_3$ | −2 | 0 | 1 | 1 | 0 | 1 | 0 | 1 |
| $q^T$ | 2 | 1 | −1 | −2 | 0 | 0 | 0 | −10 |

The most negative indicator is $q_4 = -2$. Enter $y_4$; the ratios in rows 1, 2, 3 are $3/2, 5, 1$. Thus $s_3$ leaves after step 1, raising the Phase I value from $-10$ to $-8$.

| Basic | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $s_1$ | $s_3$ | $a_2$ | RHS |
|---|---|---|---|---|---|---|---|---|
| $s_1$ | 4 | 2 | −4 | 0 | 1 | −2 | 0 | 1 |
| $a_2$ | 2 | −1 | −1 | 0 | 0 | −2 | 1 | 8 |
| $y_4$ | −2 | 0 | 1 | 1 | 0 | 1 | 0 | 1 |
| $q^T$ | −2 | 1 | 1 | 0 | 0 | 2 | 0 | −8 |

Now $y_1$ enters because $q_1 = -2$. The ratios are $1/4$ in row 1 and 4 in row 2; row 3 has negative coefficient and imposes no upper bound. Thus $s_1$ leaves after step $1/4$.

| Basic | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $s_1$ | $s_3$ | $a_2$ | RHS |
|---|---|---|---|---|---|---|---|---|
| $y_1$ | 1 | $\frac{1}{2}$ | −1 | 0 | $\frac{1}{4}$ | $-\frac{1}{2}$ | 0 | $\frac{1}{4}$ |
| $a_2$ | 0 | −2 | 1 | 0 | $-\frac{1}{2}$ | −1 | 1 | $\frac{15}{2}$ |
| $y_4$ | 0 | 1 | −1 | 1 | $\frac{1}{2}$ | 0 | 0 | $\frac{3}{2}$ |
| $q^T$ | 0 | 2 | −1 | 0 | $\frac{1}{2}$ | 1 | 0 | $-\frac{15}{2}$ |

Finally $y_3$ enters, with indicator $-1$. Only row 2 has a positive coefficient, giving step 15/2. The artificial $a_2$ leaves.

| Basic | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $s_1$ | $s_3$ | $a_2$ | RHS |
|---|---|---|---|---|---|---|---|---|
| $y_1$ | 1 | $-\frac{3}{2}$ | 0 | 0 | $-\frac{1}{4}$ | $-\frac{3}{2}$ | 1 | $\frac{31}{4}$ |
| $y_3$ | 0 | $-2$ | 1 | 0 | $-\frac{1}{2}$ | $-1$ | 1 | $\frac{15}{2}$ |
| $y_4$ | 0 | $-1$ | 0 | 1 | 0 | $-1$ | 1 | 9 |
| $q^T$ | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |

The Phase I value is zero, with $a_2$ nonbasic and zero. Delete its column; retain the nonsingular basis $(y_1, y_3, y_4)$.

### 5.2 Phase II and absence of alternative optima

Recompute the objective from $(-2, 1, -5, 1, 0, 0)$; do not reuse the Phase I row.

| Basic | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $s_1$ | $s_3$ | RHS |
|---|---|---|---|---|---|---|---|
| $y_1$ | 1 | $-\frac{3}{2}$ | 0 | 0 | $-\frac{1}{4}$ | $-\frac{3}{2}$ | $\frac{31}{4}$ |
| $y_3$ | 0 | $-2$ | 1 | 0 | $-\frac{1}{2}$ | $-1$ | $\frac{15}{2}$ |
| $y_4$ | 0 | $-1$ | 0 | 1 | 0 | $-1$ | 9 |
| $q^T$ | 0 | 11 | 0 | 0 | 3 | 7 | $-44$ |

All indicators are nonnegative, so no Phase II pivot is needed:

$$y^* = (31/4, 0, 15/2, 9), \qquad b^T y^* = -44.$$

For every feasible dual, the tableau implies

$$b^T y = -44 - 11y_2 - 3s_1 - 7s_3 \le -44. \tag{4}$$

A distinct optimizer would have to make an actual feasible move keeping the objective unchanged. Equality forces $y_2 = s_1 = s_3 = 0$; the three basic equations then fix $(y_1, y_3, y_4) = (31/4, 15/2, 9)$. Thus there is no such move and the dual optimum is unique. The zero indicators belong only to basic variables; they do not alone signal alternative optima.

### 5.3 Recover the primal without a primal simplex solve

Since $y_1, y_3, y_4 > 0$, complementary slackness makes primal rows 1, 3, 4 tight:

$$-2x_2 - 2x_3 = -2, \qquad -2x_1 + x_2 + x_3 = -5, \qquad 2x_1 + 2x_2 + x_3 = 1.$$

The first gives $x_2 + x_3 = 1$. Substitution into the second gives $x_1 = 3$; the third becomes $2x_2 + x_3 = -5$. Therefore

$$x^* = (3, -6, 7), \quad Ax^* - b = (0, 11, 0, 0), \quad c^T x^* = 9 - 60 + 7 = -44.$$

The free variable $x_2 = -6$ is admissible. Both constrained primal variables are nonnegative; the unused row 2 has surplus 11. All dual column slacks are zero. Feasibility and equal objectives certify both optima. The three equations above are independent, so the primal optimum is unique as well.

# 6 Task 5(c): equality domains reveal infeasibility

Both $x_1$ and $x_2$ are free, giving two dual equalities:

$$\begin{aligned} \max \quad & -y_1 + 5y_2 + y_3 - y_4, \\ -y_1 - y_2 - y_3 &= -1, \\ -y_1 + y_3 - y_4 &= 5, \\ 2y_1 + 3y_2 + y_3 - 2y_4 \le 8, \qquad & y \ge 0. \end{aligned}$$

The first equality implies $y_1 + y_2 + y_3 = 1$, hence $-y_1 + y_3 - y_4 \le y_3 \le 1$. It cannot equal 5.

## 6.1 One Phase I pivot makes the contradiction quantitative

Normalize the first equality by $-1$, append artificials $a_1, a_2$ to the first two rows, and use slack $s_3$ in the third. Maximize $-a_1 - a_2$ with $B = (a_1, a_2, s_3)$.

| Basic | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $s_3$ | $a_1$ | $a_2$ | RHS |
|---|---|---|---|---|---|---|---|---|
| $a_1$ | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| $a_2$ | −1 | 0 | 1 | −1 | 0 | 0 | 1 | 5 |
| $s_3$ | 2 | 3 | 1 | −2 | 1 | 0 | 0 | 8 |
| $q^T$ | 0 | −1 | −2 | 1 | 0 | 0 | 0 | −6 |

Enter $y_3$, whose indicator is $-2$. The ratios are $1, 5, 8$, so $a_1$ leaves on row 1 after step 1.

| Basic | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $s_3$ | $a_1$ | $a_2$ | RHS |
|---|---|---|---|---|---|---|---|---|
| $y_3$ | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| $a_2$ | −2 | −1 | 0 | −1 | 0 | −1 | 1 | 4 |
| $s_3$ | 1 | 2 | 0 | −2 | 1 | −1 | 0 | 7 |
| $q^T$ | 2 | 1 | 0 | 1 | 0 | 2 | 0 | −4 |

All indicators are nonnegative: Phase I stops at $z_{\mathrm{I}} = -4$, hence $w = a_1 + a_2 = 4$. Indeed its second equation reads

$$-2y_1 - y_2 - y_4 - a_1 + a_2 = 4,$$

so

$$a_1 + a_2 = 4 + 2y_1 + y_2 + y_4 + 2a_1 \ge 4.$$

This supplies a hand certificate of the Phase I optimum and the original dual's infeasibility. There is no valid Phase II or recovered optimal primal.

The primal nevertheless has the feasible ray

$$x(t) = (-5 - t, -t, 0), \qquad t \ge 0.$$

Its row values are $(5 + 2t, 5 + t, 5, t)$, dominating $(-1, 5, 1, -1)$, and its objective is $5 - 4t \to -\infty$. Therefore the primal is unbounded below; there are no dual optimal alternatives to discuss.

# 7 Task 5(d): a feasible dual, two Phase II pivots

Here only $x_2$ is free. The dual is

$$\begin{aligned} \max \quad & y_1 + 5y_2 + y_3 - 2y_4, \\ 3y_1 - y_2 - y_3 + 3y_4 &\le 1, \\ y_1 - y_3 - y_4 &= -5, \\ 2y_1 + y_2 + y_3 - 2y_4 &\le 8, \qquad y \ge 0. \end{aligned}$$

Normalize the equality to $-y_1 + y_3 + y_4 = 5$ before adding artificial $a_2$; use slacks $s_1, s_3$. The objective remains the original dual maximum.

## 7.1 Phase I

Start at $B = (s_1, a_2, s_3)$ and maximize $-a_2$.

| Basic | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $s_1$ | $s_3$ | $a_2$ | RHS |
|---|---|---|---|---|---|---|---|---|
| $s_1$ | 3 | −1 | −1 | 3 | 1 | 0 | 0 | 1 |
| $a_2$ | −1 | 0 | 1 | 1 | 0 | 0 | 1 | 5 |
| $s_3$ | 2 | 1 | 1 | −2 | 0 | 1 | 0 | 8 |
| $q^T$ | 1 | 0 | −1 | −1 | 0 | 0 | 0 | −5 |

Indicators $q_3 = q_4 = -1$ tie; take $y_3$ by its smaller index. Rows 2 and 3 have ratios 5 and 8; row 1's negative coefficient imposes no upper bound. Thus $a_2$ leaves after step 5.

| Basic | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $s_1$ | $s_3$ | $a_2$ | RHS |
|---|---|---|---|---|---|---|---|---|
| $s_1$ | 2 | −1 | 0 | 4 | 1 | 0 | 1 | 6 |
| $y_3$ | −1 | 0 | 1 | 1 | 0 | 0 | 1 | 5 |
| $s_3$ | 3 | 1 | 0 | −3 | 0 | 1 | −1 | 3 |
| $q^T$ | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |

The Phase I value is zero. Delete the nonbasic artificial. The restored objective coefficients are $(1, 5, 1, -2, 0, 0)$.

## 7.2 Phase II

| Basic | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $s_1$ | $s_3$ | RHS |
|---|---|---|---|---|---|---|---|
| $s_1$ | 2 | −1 | 0 | 4 | 1 | 0 | 6 |
| $y_3$ | −1 | 0 | 1 | 1 | 0 | 0 | 5 |
| $s_3$ | 3 | 1 | 0 | −3 | 0 | 1 | 3 |
| $q^T$ | −2 | −5 | 0 | 3 | 0 | 0 | 5 |

Enter $y_2$, with indicator $-5$. Only row 3 has a positive coefficient; $s_3$ leaves after step 3 and the objective rises from 5 to 20.

| Basic | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $s_1$ | $s_3$ | RHS |
|---|---|---|---|---|---|---|---|
| $s_1$ | 5 | 0 | 0 | 1 | 1 | 1 | 9 |
| $y_3$ | −1 | 0 | 1 | 1 | 0 | 0 | 5 |
| $y_2$ | 3 | 1 | 0 | −3 | 0 | 1 | 3 |
| $q^T$ | 13 | 0 | 0 | −12 | 0 | 5 | 20 |

Now enter $y_4$, with indicator $-12$. The eligible ratios in rows 1 and 2 are 9 and 5, so $y_3$ leaves after step 5. The objective rises by $12 \cdot 5 = 60$.

| Basic | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $s_1$ | $s_3$ | RHS |
|---|---|---|---|---|---|---|---|
| $s_1$ | 6 | 0 | $-1$ | 0 | 1 | 1 | 4 |
| $y_4$ | $-1$ | 0 | 1 | 1 | 0 | 0 | 5 |
| $y_2$ | 0 | 1 | 3 | 0 | 0 | 1 | 18 |
| $q^T$ | 1 | 0 | 12 | 0 | 0 | 5 | 80 |

This yields

$$y^* = (0, 18, 0, 5), \qquad b^T y^* = 90 - 10 = 80.$$

The final identity

$$b^T y = 80 - y_1 - 12y_3 - 5s_3 \leq 80 \tag{5}$$

forces $y_1 = y_3 = s_3 = 0$ at every optimum. The remaining equations give $y_4 = 5$, $y_2 = 18$, $s_1 = 4$. No nonzero feasible displacement can keep the objective at 80, so the dual optimum is unique.

### 7.3 Primal recovery and the role of the free variable

The dual column slacks are

$$c - A^T y^* = (4, 0, 0).$$

Because $x_1 \geq 0$, complementarity gives $x_1 = 0$. Positive $y_2, y_4$ make primal rows 2 and 4 tight:

$$-x_1 + x_3 = 5, \qquad 3x_1 - x_2 - 2x_3 = -2.$$

Hence $x_3 = 5$, $x_2 = -8$. The recovered point is

$$x^* = (0, -8, 5), \qquad Ax^* - b = (1, 0, 12, 0), \qquad c^T x^* = 40 + 40 = 80.$$

It obeys the domains, both inactive inequalities have nonnegative surplus, and the primal and dual values agree. These equations force the same point for every primal optimum, proving its uniqueness without a primal simplex run.

## 8 Interpretation and common mistakes

- **Objective signs:** task 4 negates the primal objective for its dual-simplex tableau; report 7 as the original minimum, not $-7$. Tasks 5 maximize the dual directly, including the valid negative maximum $-44$ in 5(b).
- **Free and nonpositive domains:** a free primal coordinate creates an equality in the dual. A nonpositive coordinate reverses the column inequality. Bounds on the right side do not determine these signs.
- **Two-phase transition:** artificials are feasibility devices. Delete their columns at a zero Phase I optimum and recompute the original objective. A negative Phase I maximum forbids Phase II.
- **Alternative optima:** a zero indicator is useful only with an actual feasible move. In 5(b) and 5(d), every nonbasic indicator is strictly positive, and the basic equations leave no optimal movement.
- **Primal recovery:** solve the equations forced by a dual optimizer, then verify all remaining inequalities and domains. Complementarity alone does not establish feasibility.
- **Unboundedness:** give a feasible starting point and an improving ray. An infeasible dual alone would not prove primal unboundedness if primal feasibility had not also been established.

# 9 MATLAB comparison

The companion sheet_01_dual_simplex_and_recovery.m uses the same printed matrices, augmented equations, basis orders, and pivot rules as the hand solutions. From the repository root, run

```
run('matlab/exercises/sheet_01_dual_simplex_and_recovery.m')
```

It uses base MATLAB linear solves, with absolute tolerance $10^{-10}$ for residuals, tableau comparisons, sign tests, and objective values. This tolerance is appropriate to these small inputs; it is not a conditioning guarantee for arbitrary models. The script checks the specified paths, including objective-row recomputation, independently against new basis solves. It checks primal/dual feasibility, complementarity, positive nonbasic indicators for uniqueness, and feasible rays for the unbounded cases. It does not implement a general LP solver.

| Task | Primal outcome | Dual outcome | Phase I artificial sum |
|---|---|---|---|
| 4 | Unique minimum 7 | Unique maximum 7 | Not used |
| 5(a) | Unbounded below | Infeasible | 3 |
| 5(b) | Unique minimum $-44$ | Unique maximum $-44$ | 0 |
| 5(c) | Unbounded below | Infeasible | 4 |
| 5(d) | Unique minimum 80 | Unique maximum 80 | 0 |

The table records exact expected hand results, not invented solver output. Mathematical, static-analysis, and MATLAB execution evidence is identified separately in the final remarks and repository verification record.

# 10 Sources and editorial remarks

**Task source:** ../execrices-2026-27/ProgMatc1.pdf, PDF page 2, tasks 4 and 5(a)–5(d). The Polish instructions and mathematical data are preserved in Section 1; layout is retypeset. The supplied current-course context is 2026–27. The PDF creation metadata is 2025, which does not establish the delivery date. No lecture date is claimed.

The hand derivations, complete tableaus, recovery equations, rays, and uniqueness arguments are explanatory additions. The generic instruction in task 5 requests optimal solutions. Sections 4 and 6 show that the printed domains and coefficients instead imply unbounded primal problems and infeasible duals. The task statements remain unchanged; their intended variants are unknown. Successful Phase II solves are provided for 5(b) and 5(d).

**Verification status:** hand tableaus and pivot choices were independently recomputed using exact rational arithmetic. MATLAB execution is initially unverified; a successful static check is not MATLAB execution. The companion's actual execution evidence and release, when available, belong in docs/matlab-verification.md.